# Pyung Kim

Postdoctoral Scholar

Center for Spatial Studies and Data Science
Department of Geography
University of California, Santa Barbara
(Ellison Hall, Office 5841)

pyungkim@ucsb.edu | https://pyungkim.github.io | Google Scholar

## ACADEMIC APPOINTMENTS

Postdoctoral Scholar 2026 – Present
*Center for Spatial Studies and Data Science, Department of Geography, University of California, Santa Barbara*

## PROFESSIONAL AFFILIATIONS

Research Affiliate 2024 – 2025
*Spatial Health AI Research Partnership (SHARP), The University of Texas at Dallas*

Instructor 2023 – 2025
*Korea Social Science Data Archive (KOSSDA), Seoul National University*

Research Assistant 2020
*Korea Institute of Public Administration (KIPA)*

## EDUCATION

Ph.D. in Public Policy and Political Economy 2025
*The University of Texas at Dallas* (Brian J.L. Berry Scholarship, Highest Honor)
Dissertation: *Long-Term Effects of Neighborhoods on Educational Attainment by Disability Status*

M.S. in Geospatial Information Science 2025
*The University of Texas at Dallas*

M.A. in Public Administration 2020
*Yonsei University* (Brain Korea 21 PLUS Scholarship, National Research Foundation)

B.A. in Public Administration 2018
*Ajou University* (National Excellence Scholarship, Deputy Prime Minister of Korea)
*Julius-Maximilian University of Würzburg* (Exchange Student, Germany, 2016–2017)

## RESEARCH AREAS

| | |
|---|---|
| **Substantive** | Public Policy, Public Health, Human Geography, Neighborhood Effects, Disability |
| **Methodological** | GIS, Spatial Data Science, Causal Inference, Quasi-experiment, Machine Learning |

## PUBLICATIONS *[*] Corresponding author, [†] Student co-author*

5. **Kim, P.**, Choi, S.[†], Kim, D., & Lee, C[*]. (2026). Bayesian spatio-temporal modeling for policy evaluation: Sensitivity of policy effect estimates in the context of COVID-19 stay-at-home orders. *PLoS One 21(2): e0339196.*

4. **Kim, P.** [*]**,** Kim D., Scotch R., Jeong D.[†], & Kowalske K. (2025). Geospatial analysis of community-level social and environmental barriers for adult burn injury survivors in North Texas. *Burns*, 51(5), 107512.

3. Hong, S.[*], Jeong, D.[†], & **Kim, P.** (2024). Have offender demographics changed since the COVID-19 Pandemic? Evidence from money mules in South Korea**.** *Journal of Criminal Justice*, 91, 102156.

2. Maxwell, S. P., Brooks, C., **Kim, P.**, Kim, D., McNeely, C. L., & Thomas, K[*]. (2023). Understanding habitats and environmental conditions of white-tailed deer population density and public health data to aid in assessing human tick-borne disease risk. *Microorganisms*, 11(4), 865.

1. **Kim, P.** & Bae, H.[*] (2022). Do firms respond differently to the carbon pricing by industrial sector? How and why? A comparison between manufacturing and electricity generation sectors using firm-level panel data in Korea. *Energy Policy*, 162, 112773.

## UNDER REVIEW

**Kim, P.** [*] "From Command-and-Control to Carbon Pricing: Estimating the Causal Effects on Firm Environmental Performance" (Available at SSRN)

**Kim, P.** [*] "Neighborhood effects on educational attainment by disability status"

Kim, H.[†], **Kim, P.**[*], "Modeling Policy Diffusion Across Space and Time: A Bayesian Spatial–Temporal Approach"

**Kim, P.**, Jung, J. [†], Min, C., Choi, H., Kim., D, Kumar, P. [†], Hwang, J., & Song, D[*]. "mHealth Intervention Integrating Personal PM2.5 Monitoring and Deep Learning to Reduce Pediatric Asthma Exacerbations: A Pilot Study"

## WORKING PAPERS

"Geospatial Analysis on Dental Health Service Access and Workforce Distribution in Kenya" (with Brenda Okumu, Caroline Kibosia, Dohyeong Kim, Arthur Kemoli, Osman Adullahi, & Hyewon Lee)

"The Impact of Medicaid Expansion on Cancer Outcomes: Evidence on the Moderating Role of Oncology Care Access" (with Kristine Hahm)

"Administrative Deserts in the U.S. Social Security: Field Office Accessibility and SSI Participation" (with Peter Kedron)

"When Does Moving Matter? Residential Relocation and Academic Outcomes Among Children" (with Peter Kedron)

## EXTERNAL GRANTS *Non-U.S. awards are reported in USD using award-date exchange rates.*

*I have participated in awarded grants totaling $3,717,126, of which my recognized contribution was $32,800.*

[PI / Co-PI]

1. What Is the Relationship Between Adolescent Neighborhood Environments and Long-Run Labor Market Outcomes for Children with Disabilities? 2024 – 2025
   - **PI: Pyung Kim**, UTD
   - *Social Security Administration via Retirement and Disability Research Center at UW-Madison* ($6,000; Recognition: $6,000)

[Key Personnel]

4. Interventional Study to Prevent Exacerbation of Atopic Dermatitis by Exposure to Particulate Matter 2024 – 2027
   - Role: Data Scientist
   - PI: Kangmo Ahn, *Samsung Medical Center*
   - *Korea Disease Control and Prevention Agency* ($629,630; Recognition: $4,600)

3. Geospatial Community Mapping to Assess Social and Environmental Barriers and Facilitators for Burn Survivors 2022 – 2026
   - Role: GIS Specialist
   - PI: Karen Kowalske, *UTSW Medical Center*; co-PI: Dohyeong Kim & Richard Scotch, *UTD*
   - *National Institute on Disability, Independent Living, and Rehabilitation Research* ($2,320,495; Recognition: $12,600)

2. The Capacity-Building Project for Primary Healthcare Workforce to Improve Maternal and Child Oral Health in Kilifi County, Kenya 2025 – 2026
   - Role: GIS Specialist **(field study in Kenya, October 2025)**
   - PI: Hyewon Lee, *Seoul National University*
   - *Korea International Cooperation Agency* ($440,000; Recognition: $5,600)

1. Spatial Distribution of Natural Radiation in Jeonbuk and Jeonnam Provinces 2023 – 2025
   - Role: GIS Specialist
   - PI: Yeol Rho, *Chonnam National Univ.*; co-PI: Dohyeong Kim, *UTD*
   - *National Institute of Environmental Research of Korea* ($320,000; Recognition: $4,000)

## AWARDS AND HONORS

[External Awards]

Public Administration Research Conference Travel Award, *University of Central Florida* 2025

Junior Scholar Intensive Training (JSIT), *Social Security Administration via UW-Madison* 2024

President Travel Award for APPAM, *Korean Association for Policy Studies* 2023

Brain Korea 21 PLUS Scholarship, *National Research Foundation* 2018 – 2020

The Next Century Humanities Scholarship (National Excellence Scholarship), 2015 – 2018
*awarded by the Deputy Prime Minister; funded by the Korea Student Aid Foundation*

[Internal Awards]

| | |
|---|---|
| GIS Pioneer Research Scholarship, *UT Dallas & Pioneer Natural Resources* | 2025 |
| Keith Lankford Taylor Fellowship, *UT Dallas* | 2023 |
| Brian J.L. Berry Scholarship (Highest Honor), *UT Dallas* | 2021 – 2025 |
| UT Dallas Graduate Studies Scholarship, *UT Dalla* | 2021 – 2025 |
| YUPA Graduate Studies Scholarship, *Yonsei University* | 2018 – 2020 |
| President's Award for Outstanding Graduate, *Ajou University* | 2018 |
| Teaching Assistantship for Undergraduate Student, *Ajou University* | 2015 – 2018 |
| Excellence in Undergraduate Research Project, *Ajou University* | 2015 |
| Merit-based Full Tuition Scholarship, *Ajou University* | 2014 |
| Dean's list, *Ajou University* | 2014; 2015 |

## MEDIA APPEARANCES

UT Dallas News. (2026). *How Communities Shape Burn Survivors' Return to Society*. https://news.utdallas.edu/social-sciences/burn-survivors-societal-reentry-2026/

## INVITED TALKS

"Long-Term Effects of Neighborhoods on Educational Attainment by Disability Status", *Geography Colloquium, University of California, Santa Barbara* — April 2026 (planned)

"Using GIS to Improve Rural Health Service Planning: Lessons from Kilifi County, Kenya", *Kilifi Department of Health in Kilifi County, Kenya* — Oct 2025

## TEACHING

[Co-Instructor]

Machine Learning and Deep Learning for Social Science (with R) — Winter 2026
- *Korea Social Science Data Archive* (graduate)
- *Rating: 9.64/10*

GIS and Spatial Statistics (with QGIS, GeoDa, and R) — Summer 2025
- *Korea Social Science Data Archive* (graduate)
- *Rating: 9.91/10*

Advanced GIS and Spatial Statistics (with R) — Summer 2023
- *Korea Social Science Data Archive* (graduate)
- *Rating: 9.84/10*

Introductory GIS and Spatial Statistics (with QGIS, GeoDa, and STATA) — Summer 2023
- *Korea Social Science Data Archive* (graduate)
- *Rating: 9.54/10*

[Guest Lecture]

- Quasi-Experimental Mover Designs, Geography and Policy Lab, *UC Santa Barbara* Feb 2026
- Analysis of Variance, undergraduate statistics course (EPPS 2302), *UT Dallas* April 2025
- Probability Distributions, undergraduate statistics course (EPPS 2302), *UT Dallas* Feb 2025
- Introductory Calculus, graduate statistics course (EPPS 7313), *UT Dallas* Sep 2023

## CONFERENCE PRESENTATIONS

* APPAM (Association for Public Policy Analysis and Management); APHA (American Public Health Association); ASPA (American Society for Public Administration); ACRM (American Congress of Rehabilitation Medicine); SER (Society for Epidemiologic Research); ASA (American Sociological Association)

2025 APPAM, APHA, ASA, SER
2024 APPAM, ACRM, ASPA
2023 APPAM, APHA, ASPA

## TECHNICAL SKILLS

Statistical Analysis (R, STATA)
Geospatial Analysis (R, Arc GIS, QGIS, GeoDa, Tableau)
Machine Learning / Deep Learning (R)

## RESTRICTED DATA ACCESS

2. The Impacts of School Neighborhoods on the Academic and Labor Market Outcomes of Students with Disabilities 2024
   - *PI:* ***Pyung Kim***
   - *Student-level longitudinal data (1999 – 2023, N $\geq$ 5,000,000)*
   - *Source: Texas Schools Project*

1. Geospatial Community Mapping to Assess Social and Environmental Barriers and Facilitators for Burn Survivors 2022
   - *PI: Karen Kowalske, UTSW Medical Center, UTD*
   - *Patient-level longitudinal data (1994 – 2022, N $\geq$ 25,000)*
   - *Source: National Institute on Disability, Independent Living, and Rehabilitation Research*

## SERVICE

[Editorial Board Member]

*Journal of Safety and Crisis Management* 2026 – Present

[Peer Reviewer]

*Policy Studies Journal; Scientific Reports; Energy Policy; Energy Economics; International Journal of Finance and Economics; PLOS ONE*

[University Service]

Secretary, *Korean Student Association, UT Dallas* 2023 – 2024
Vice President, *Graduate Student Assembly in Public Administration, Yonsei University* 2018 – 2019
Student Mentor, *Office of Admissions, Ajou University* 2014 – 2015